# Wadim **Müller**

Embedded Software Expert · Linux Enthusiast

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*"He who stops being better stops being good." (Oliver Cromwell)*

## Education

### Technische Hochschule Aschaffenburg (University of Applied Sciences)

*Aschaffenburg, Germany*

B.ENG. IN ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY

*Sep. 2008 - Aug. 2012*

- Emphasis on Microelectronics with final grade of 1.5 (First Class Honours).
- Thesis: Development of an USB Device Interface based on the Test and Measurement USB Class Standard (USB TMC).

### Hochschule Darmstadt (University of Applied Sciences)

*Darmstadt, Germany*

M.SC. IN ELECTRICAL ENGINEERING

*Sep. 2014 - Aug. 2017*

- Extra-occupational study with emphasis on Microelectronics with final grade of 1.1 (First Class Honours).
- Thesis: Development of an FPGA accelarated Linux System for Realtime Video Feature extraction using OpenCV on a Zynq Multicore pSoC. https://github.com/wafgo/morphological-filter-ip-core, https://github.com/wafgo/disparity-coprocessor-ip-core, https://github.com/wafgo/linux-kernel-custom-hw-uio, https://github.com/wafgo/rt-depth-map.

## Skills

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|---|---|
| **Programming** | C, C++, Python, Perl, Bash |
| **Virtualization/Emulation** | QEMU, KVM |
| **Debugging** | Lauterbach, Trace32, GDB, OpenOCD, Greenhills MULTI, ARM Core Sight |
| **Development Tooling** | CMake, GNU make, kbuild, bitbake, gcc, binutils, llvm/clang, ghs, capstone, pyspark |
| **Languages** | German, English, Russian |

## Experience

### CMBlu Energy AG

*Alzenau, Germany*

Embedded Linux Developer

*since Feb. 2025*

- Developed a custom Yocto Linux distribution for a Redox-Flow Battery Management System based on TI AM64xx SoC architecture.
- Performed complete board bring-up and hardware validation for custom AM64xx PCB designs.
- Designed and integrated Linux kernel drivers for input capture, PWM control, and rotary encoder interfaces.
- Implemented QEMU-based hardware emulation environment with custom kernel drivers (Counter/PWM with sysfs interfaces) to enable comprehensive application software testing and driver validation without physical hardware dependencies.

### Continental Automotive GmbH

*Babenhausen, Germany*

Software System Integrator/Embedded Linux Developer

*Jul. 2018 - Jan. 2025*

- Integrated Hardware-Vendor-provided Board Support Packages into Instrument Cluster Products based on Renesas Multicore ARMv7/v8 targets running Greenhills INTEGRITY OS.
- Designed and deployed Build Systems for various Instrument Cluster Products based on CMake.
- Debugged system level errors on Multicore ARMv8/v7 Systems using JTAG with GHS (MULTI) and Lauterbach Tools (Trace32).
- Implemented several RCar3 SoC Device Emulations in QEMU. Deployed machine emulation of customized Renesas RCar3 boards using QEMU https://github.com/wafgo/qemu/tree/rcar3_port.
- Deployed a Yocto based Linux Distribution as a Flash-Frontend for Instrument Clusters running Greenhills INTEGRITY OS.
- Established a core architecture for regulating of outbound DNS traffic in multi-account and multi-VPC environments utilizing AWS Route53 DNS Firewall and FMS. This significantly increased the level of security confidence in the financial sector's segregated environment.
- Developed and Deployed Debugging Tools for GHS based systems to support low level error detection (offline stackdump unwinding on armv8, ARM ETM Setup etc.)

### Magna Powertrain

*Sailauf, Germany*

EMBEDDED SOFTWARE ENGINEER

*Jan. 2017 - Jul. 2018*

- Implemented Algorithms for Brushless DC Drive Controls on resource limited Microcontrollers.
- Ported FreeRTOS to the NXP S12Z CPU Architecture.
- Designed a Software Architecture to support the development of MCU agnostic drive control algorithms including kbuild style buildsystem based on GNU make.

**Rohde & Schwarz** *Mainhausen, Germany*

Embedded Software Engineer *Jan. 2009 - Dec. 2016*

- Implemented several Linux Kernel Drivers for customized Freescale IMX6 based Boards used in Rohde & Schwarz Test and Measurement Devices.
- Developed Bringup Software for Freescale IMX6 and Renesas SH2A Boards.
- Developed a SPI to Ethernet bridge based on an PIC32 MCU with a MCU Firmware and the corresponding Linux Kernel Network driver on the Renesas SH7264 (SH2A).
- Developed and Deployed a Lauterbach Trace32 Debug Script Infrastructure to support early boot debugging.
- Implemented an USB Device Interface based on the TMC (Test and Measurement Standard) which is used in several Rohde & Schwarz Measurement devices for configuration and data retrieval over USB.
- Implemented many Application Level Modules (including corresponding Unit Tests) for several Test and Measurement Devices (LCR/Multimeter/H-Bridge).